AE4870A Rocket Motion — Assignment 2

Airborne Launch to LEO

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(a) Vis-Viva Equation

The vis-viva equation, derived from the conservation of mechanical energy [1] in a 2 body gravitational system, relates the instantaneous orbital speed v of a spacecraft to its radial distance from Earth's centre r and the semi-major axis a of the Keplerian orbit. The vis-viva equation is given by:

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right) \tag{1}$$

v	instantaneous orbital speed	$[\mathrm{m\,s^{-1}}]$
μ	standard gravitational parameter	$[\mathrm{m^3\,s^{-2}}]$
r	radial distance from Earth's centre	[m]
a	semi-major axis of the orbit	[m]

The specific orbital energy ε $[\mathrm{J\,kg^{-1}}]$ is conserved, therefore it can be written as the sum of kinetic energy $(\frac{v^2}{2})$ and gravitational potential energy $(-\frac{\mu}{r})$:

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} \tag{2}$$

Substituting Equation 1 in Equation 3 we get:

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} \tag{3}$$

For a **circular** orbit at altitude h, $r = a = R_E + h$, giving:

$$v_{\mathrm{orb}} = \sqrt{\mu/r} \tag{4}$$

$$\varepsilon_{\mathrm{orb}} = -\mu/(2r) \tag{5}$$

Computing for the target orbit at $h = 185$ km:

$$r_{\mathrm{t_{orb}}} = R_E + h = 6{,}371{,}000 + 185{,}000 = 6{,}556{,}000\ \mathrm{m} \tag{6}$$

$$v_{\mathrm{t_{orb}}} = \sqrt{\frac{\mu}{r_{\mathrm{t_{orb}}}}} = 7{,}797\ \mathrm{m\,s^{-1}} \tag{7}$$

$$\varepsilon_{\mathrm{t_{orb}}} = -\frac{\mu}{2r_{\mathrm{t_{orb}}}} = -3.040 \times 10^7\ \mathrm{J\,kg^{-1}} \tag{8}$$

The negative sign indicates that the orbit is bounded.

(b) ΔV Gain Calculation

To determine the ΔV gain for each of the 16 air-launch cases, the vis-viva equation is applied under a simplified framework with the following assumptions:

- Two-body gravitational model
- No aerodynamic drag or thrust losses
- Instantaneous velocity change

The specific energy of the reference case i.e., the ground launch at rest ($h_0 = 0$, $V_0 = 0$) is:

$$\varepsilon_{\mathrm{ground}} = -\frac{\mu}{R_E} = -6.257 \times 10^7\ \mathrm{J\,kg^{-1}} \tag{9}$$

[1] the sum of kinetic and gravitational potential energy remains constant at every point along the trajectory.

The ΔV to reach the target orbit from the ground is:

$$\Delta V_{\mathrm{ground}} = \sqrt{2\left|\varepsilon_{\mathrm{orb}} - \varepsilon_{\mathrm{ground}}\right|} = 8021\ \mathrm{m\,s^{-1}} \tag{10}$$

The ΔV gain for each air-launch case at altitude h_0 and Mach number M can be written in terms of two independent contributions:

$$\Delta V_{\mathrm{gain}} = \Delta V_{\mathrm{pot}} + V_0 \tag{11}$$

The first contribution, V_0, is the launch velocity imparted directly by the carrier vehicle at ignition. It can be expressed as:

$$V_0 = M \cdot a(h_0) \tag{12}$$

where $a(h_0)$ is the speed of sound at launch altitude h_0, obtained from the International Standard Atmosphere (ISA) model. The ISA defines the atmosphere's temperature profile $T(h_0)$ through Equation 14, from which the speed of sound follows via Equation 13.

$$a(h_0) = \sqrt{\gamma R T(h_0)} \tag{13}$$

$$T(h_0) = \begin{cases} 288.15 - 0.0065\,h_0 & h_0 < 11{,}000\ \mathrm{m} \\ 216.65 & 11{,}000 \le h_0 \le 20{,}000\ \mathrm{m} \end{cases} \tag{14}$$

To evaluate the ISA at arbitrary altitudes programmatically, the `ambiance` Python module is used.

For the ground reference case, we take $V_0 = 0$. The second contribution, ΔV_{pot}, is the velocity gain arising from the gravitational potential energy at launch altitude.

$$\Delta V_{\mathrm{pot}} = \Delta V_{\mathrm{ground}} - \sqrt{2\left|\varepsilon_{\mathrm{orb}} + \frac{\mu}{R_E + h_0}\right|} \tag{15}$$

Substituting Equations 15 and 12 into Equation 11, the ΔV gain for each air-launch case is expressed as:

$$\Delta V_{\mathrm{gain}} = \Delta V_{\mathrm{ground}} - \sqrt{2\left|\varepsilon_{\mathrm{orb}} + \frac{\mu}{R_E + h_0}\right|} + M \cdot a(h_0) \tag{16}$$

Table 1: ΔV_{gain} [m/s] vs. ground launch.

h_0	M = 0	M = 1	M = 2	M = 3
0 km	0.0	340.3	680.6	**1020.9**
7.5 km	9.2	319.4	629.6	939.8
15 km	18.3	313.4	608.5	903.6
22.5 km	27.5	324.2	621.0	917.6

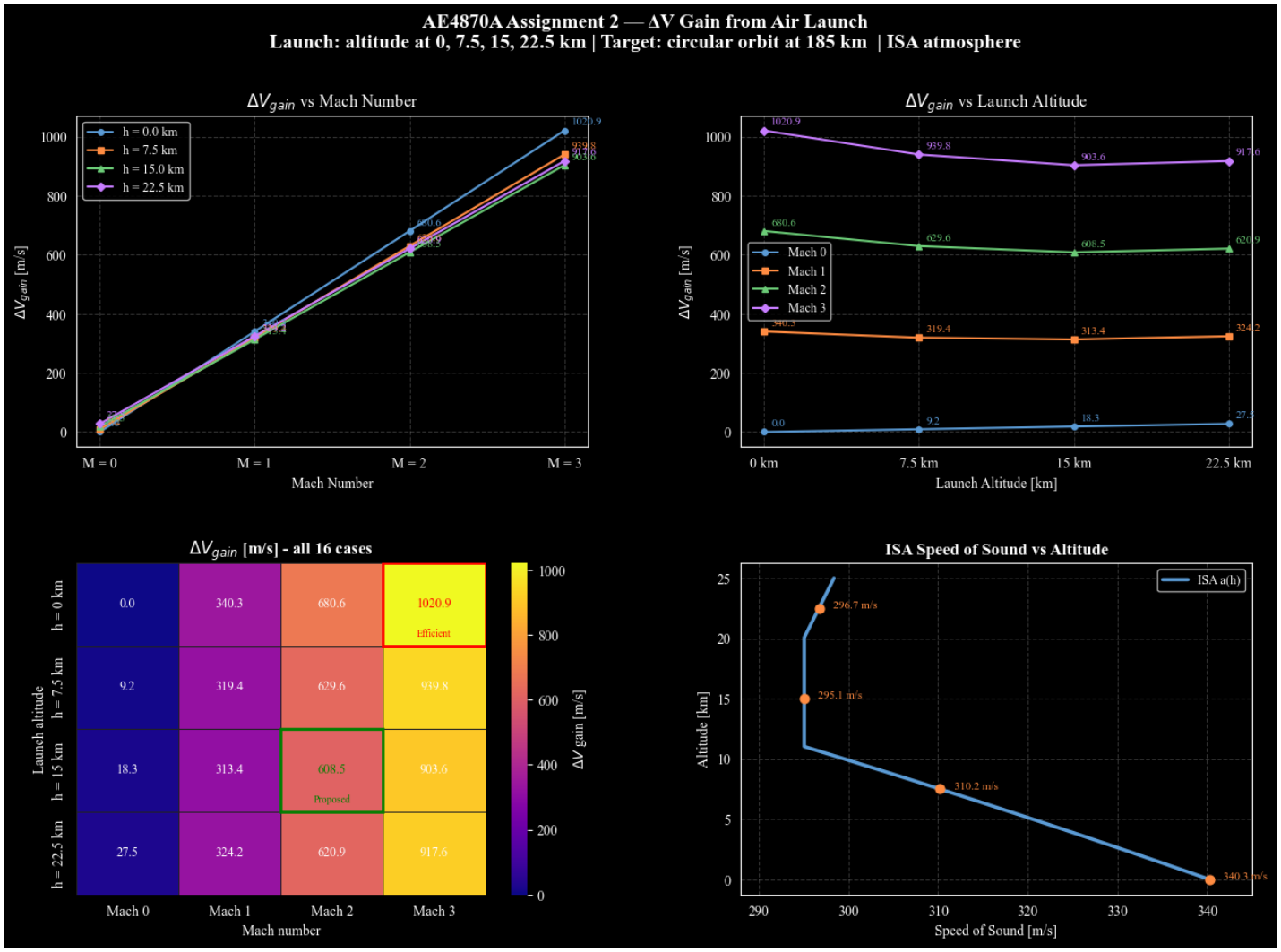

Figure 1: ΔV gains for all 16 cases. Most efficient value: $h_0 = 22.5\,\text{km}$, Mach 3 ($\Delta V_{\text{gain}} = 77.2\,\text{m/s}$). Proposed value: $h_0 = 0\,\text{km}$, Mach 3 ($\Delta V_{\text{gain}} = 1020.9\,m/s$)

(c) Comparison with Figure 1A

Table 2 compares the computed ΔV gains against the values reported in Figure 1a of [1] for a horizontal launch ($\gamma = 0°$).

Table 2: Comparison of computed ΔV_{gain} [m/s] against Figure 1A of [1] at $\gamma = 0°$.

	M = 0	M = 1	M = 2	M = 3
		Computed		
0 km	0.0	340.3	680.6	1020.9
7.5 km	9.2	319.4	629.6	939.8
15 km	18.3	313.4	608.5	903.6
22.5 km	27.5	324.2	621.0	917.6
		Paper (Fig. 1A, $\gamma = 0°$)		
0 km	∼0	∼340	∼680	∼1020
7.5 km	∼50	∼400	∼750	∼1100
15 km	∼200	∼600	∼900	∼1200
22.5 km	∼400	∼800	∼1100	∼1400
		Difference		
0 km	0	0	0	0
7.5 km	41	81	120	160
15 km	182	287	292	296
22.5 km	373	476	479	482

As seen in Table 2, there is a clear discrepancy between the gains computed and those reported in [1]. The underlying reason lies in the methods utilized. The paper [1] employs Program to Optimize Simulated Trajectories (POST), a 3 DOF optimization code that accounts for various losses such as steering, gravity, drag and atmospheric pressure losses. Air launches result in the reduction of each of these losses simultaneously, which is not accounted for in the energy method (which only considers kinetic and potential energy contributions) applied here.

At low altitudes and Mach 0, trajectory losses are small indicating that the energy method effectively captures the launch dynamics under these conditions. However, as Mach number and altitude increase, the curves diverge, since the loss terms are functions of both altitude and velocity. Despite these discrepancies in magnitude, both methods exhibit similar trends in ΔV_{gain}. Thus, the initial launch velocity can be identified as the dominant gain parameter.

From the comparisons made, it can be concluded that the energy method is suitable only for first-order estimates of ΔV_{gain}, whereas accurate predictions require the incorporation of trajectory-dependent effects, such as steering, gravity, and drag losses.

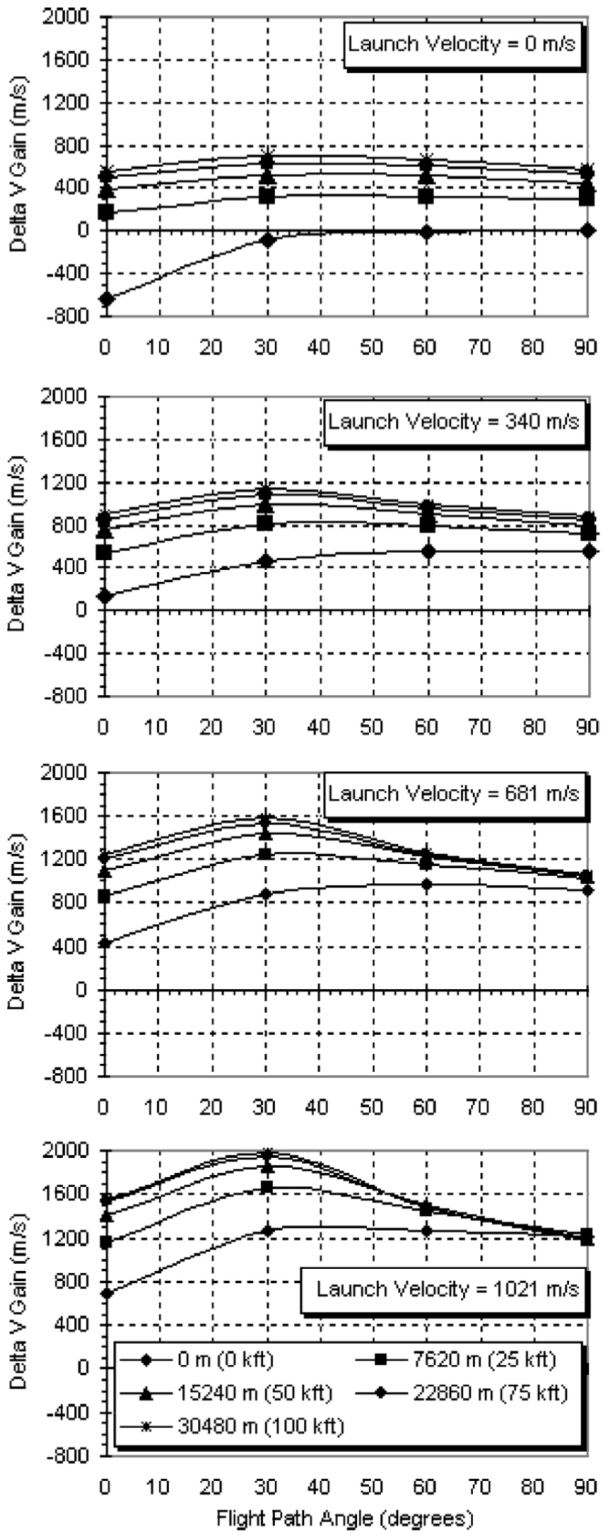

Fig. 1a Delta-V gain (no wing).

Figure 2: Figure 2: Results from reference paper (Figure 1a).

(d) Effect of Losses and Earth Rotation

The full ΔV budget for a ground-launched vehicle reaching LEO [1] is given by:

$$\Delta V_{\text{ideal}} = V_{\text{orbit}} + \Delta V_{\text{drag}} + \Delta V_{\text{gravity}} + \Delta V_{\text{steering}} + \Delta V_{\text{pressure}} - V_{\oplus} - V_{\text{carrier}} \tag{17}$$

The magnitudes of the loss terms in Equation 17 are dependent on the launch conditions and are each affected differently when switching from a conventional ground launch to an air launch.

Gravity losses arise due to the work done by the rocket engine against Earth's gravitational field during ascent. For a conventional ground launch, gravity losses are typically in the range of 1150-1600 m/s [1] and expressed as:

$$\Delta V_{\text{gravity}} = \int_0^{t_{\text{bo}}} g(r)\, \sin\gamma(t)\, \mathrm{d}t \tag{18}$$

where $g(r) = \mu/r^2$ is the gravitational acceleration at radial distance r, $\gamma(t)$ is the flight-path angle at time t, and t_{bo} is the burnout time.

Their high range makes them the dominant loss term in the ΔV budget. With air launching, this effect is reduced as altitude increases, atmospheric density decreases, aerodynamic drag decreases and the vehicle accelerates faster. Consequently, the rocket spends less time thrusting against gravity, resulting in the reduction of gravity losses. Additionally, the flight path angle contributes significantly to gravity losses. A steeper ascent results in increased losses as$\Delta V_{\text{gravity}} \propto \sin\gamma(t)$. However, an exact estimate can't be obtained as the flight path angle is determined by POST trajectory optimization and is not explicitly available for evaluation in this analysis. Consequently, gravity losses can't be quantified for the energy-based framework we are considering,

Drag losses arise due to aerodynamic drag forces acting on the vehicle during ascent through the atmosphere. For a conventional ground launch, drag losses are typically in the range of 40-160 m/s [1] and are expressed as:

$$\Delta V_{\text{drag}} = \int_0^{t_{\text{bo}}} \frac{D(t)}{m(t)}\, \mathrm{d}t \tag{19}$$

where $D(t)$ is the aerodynamic drag force at time t and $m(t)$ is the instantaneous vehicle mass. The drag force is given by:

$$D = \frac{1}{2}\, \rho(h)\, v^2\, C_D\, A \tag{20}$$

where $\rho(h)$ is the ambient air density at altitude h, v is the vehicle speed, C_D is the drag coefficient, and A is the reference area.

Since $\rho(h)$ decreases with altitude as defined by the ISA model, air launching at 7.5-22.5 km results in an ambient density reduction of 40-90% relative to sea level. As density directly governs the drag force through Equation 20, reduction in $\rho(h)$ decreases D proportionally, which reduces ΔV_{drag}.

Earth's rotation about its axis imparts an additional eastward velocity to the launch vehicle. The magnitude of this contribution depends on the launch latitude, as given by Equation 21, and amounts to ∼408 m/s at 28.5° latitude. Since both ground and air launch occur from the same position, this contribution is equal in both cases and therefore cancels in the ΔV_{gain} calculation. However, air launching potentially offers a greater Earth rotation benefit than a fixed ground site as the carrier aircraft can fly to a

more favourable latitude or azimuth prior to release, increasing $V_\oplus$ beyond what a ground launch can achieve.

$$V_\oplus = \omega_E \, R_E \cos\phi \tag{21}$$

Other losses such as steering losses arise from the requirement to manoeuvre the launch vehicle to the optimal flight-path angle. Since manoeuvring is required whenever the thrust vector is misaligned with the velocity vector, air launching proves advantageous as the rocket can be released at a flight-path angle close to the optimal trajectory. This minimizes post-ignition manoeuvring and the resulting steering losses. The paper recommends a minimum flight-path angle of 30° above the horizontal at ignition to limit these losses [1]. Another loss term is atmospheric pressure loss, which arises as ambient pressure causes under-expansion or over-expansion of the nozzle exhaust, reducing thrust relative to vacuum operation. As ambient pressure decreases with altitude, the nozzle exit pressure p_e approaches the ambient pressure p_a, reducing nozzle expansion losses.

Table 3: Typical loss magnitudes, ground launch to LEO.

Loss term	[m/s]
Gravity losses	1150 – 1600
Drag losses	40 – 160
Steering losses	∼50
Pressure losses	∼50
Earth rotation gain	∼408 at 28.5°

(e) Most Realistic and Efficient Case

The most realistic and efficient air-launch case is selected by evaluating 2 criteria: efficiency and feasibility.

Efficiency is a measure of how effectively a system converts its input resources into a desired output while minimizing losses. In the context of this problem, it is defined as the case yielding the maximum ΔV_{gain} for a given launch condition.From Table 1, the highest gain is achieved at $h_0 = 0\,\text{km}$ and $M_0 = 3$, with $\Delta V_{\text{gain}} = 1020.9\,\text{m/s}$. This is consistent with the finding that ΔV_{gain} increases proportionally with both Mach number and launch velocity.

Feasibility is a measure of how practically achievable a launch condition is. In the context of this problem, it translates to the achievable velocity and altitude of the carrier aircraft. Although Mach 3 was identified as the most efficient case, it is not feasible as it would require a carrier aircraft of SR-71 class, which would significantly increase the development and operational costs. Downgrading to Mach 2 remains effective and achievable with existing supersonic military aircraft (such as F-15), while allowing for a flight path angle of 30° as recommended by [1] to limit steering losses.

Regarding altitude, the paper [1] notes that ΔV gains plateau above approximately 15 km, meaning further increases in altitude yield diminishing returns. An altitude of 15 km proves to be a suitable choice, as it not only lies within the operational range of a supersonic military aircraft but also places the rocket above the majority of the troposphere, where atmospheric density is approximately 26% of its sea-level value (evaluated from the ISA model).

Thus, $\boxed{h_0 = 15\,\text{km},\ M_0 = 2}$ represents the most efficient and feasible case.

(f) Dawn Aerospace Scenario

The ΔV gain for the Dawn Aerospace scenario is calculated using Equation 16. The launch conditions are $h_0 = 120\,\text{km}$ and $V_0 = 2000\,\text{m}\,\text{s}^{-1}$.

The radial distance at the Dawn launch altitude is:

$$r_{\text{Dawn}} = R_E + 120{,}000 = 6{,}491{,}000\ \text{m} \tag{22}$$

The specific potential energy at 120 km evaluated at $V_0 = 0$, is:

$$\varepsilon_{\text{Dawn}} = -\frac{\mu}{r_{\text{Dawn}}} = -\frac{3.986 \times 10^{14}}{6{,}491{,}000} = -6.141 \times 10^7\ \text{J}\,\text{kg}^{-1} \tag{23}$$

The ΔV from launch orbit to the target orbit is:

$$\Delta V_{\text{alt}} = \sqrt{2\,|\varepsilon_{\text{target}} - \varepsilon_{\text{Dawn}}|} = \sqrt{2 \times 3.101 \times 10^7} = 7875.5\ \text{m}\,\text{s}^{-1} \tag{24}$$

The potential energy contribution to the ΔV gain is:

$$\Delta V_{\text{pot}} = \Delta V_{\text{ground}} - \Delta V_{\text{Dawn}} = 8021 - 7875.5 = 145.5\ \text{m}\,\text{s}^{-1} \tag{25}$$

Adding the launch velocity V_0, the total ΔV gain is:

$$\boxed{\Delta V_{\text{gain}} = \Delta V_{\text{pot}} + V_0 = 145.5 + 2000 = 2145.5\ \text{m}\,\text{s}^{-1}} \tag{26}$$

$$\boxed{\frac{\Delta V_{\text{gain}}}{\Delta V_{\text{ground}}} \times 100 = \frac{2145.5}{8021} \times 100 = 26.7\%} \tag{27}$$

This represents a reduction of 26.7% relative to a conventional ground launch.

The ΔV still required from the rocket engine is:

$$\Delta V_{\text{rocket}} = \Delta V_{\text{ground}} - \Delta V_{\text{gain}} = 8021 - 2145.5 = 5875.5\ \text{m}\,\text{s}^{-1} \tag{28}$$

The large ΔV_{gain} of $2145.5\,\text{m}\,\text{s}^{-1}$ is attributed to the head start provided by the carrier aircraft's initial velocity $V_0 = 2000\,\text{m}\,\text{s}^{-1}$ and the potential energy contribution of launching at $h_0 = 120\,\text{km}$. As 120 km lies above the Karman line, the rocket ignites in a near-vacuum environment making atmospheric drag and pressure losses negligible. Also, the trajectory spans only 65 km of altitude, which significantly reduces gravity losses relative to a ground launch.

As the mass fraction can be expressed through the Tsiolkovsky rocket equation:

$$\frac{m_0}{m_f} = e^{\Delta V / v_e} \tag{29}$$

where m_0 is the initial mass, m_f is the final dry mass and $v_e = I_{\text{sp}} \cdot g_0$ is the effective exhaust velocity. A reduction of 26.7% in the required ΔV (Equation 27), results in a lower mass ratio and consequently a reduced propellant mass fraction.

The advantages and disadvantages of the Dawn Aerospace scenario are summarised in Table 4.

Table 4: Advantages and disadvantages of the Dawn Aerospace air-launch scenario ($h_0 = 120\,\mathrm{km}$, $V_0 = 2000\,\mathrm{m\,s^{-1}}$).

Advantages	Disadvantages
• ΔV_{gain} of $2145.5\,\mathrm{m\,s^{-1}}$ is the largest of all cases considered. • All atmospheric losses are eliminated as the rocket ignites above the Kármán line. • Short 65 km trajectory to target orbit reduces gravity losses. • Reduced propellant mass fraction allows for a smaller launch vehicle or greater payload capacity. • Flexible launch azimuth and inclination enabled by carrier aircraft mobility.	• Rocket still requires $\Delta V_{\text{rocket}} = 5875.5\,\mathrm{m\,s^{-1}}$ from its own propulsion system. • Carrier vehicle experiences extreme thermal and structural loads during ascent. • Dawn Aerospace remains in early development phases and is less feasible due to high development and operational costs of a sub-orbital spaceplane.

(g) Proposed Scenario

The ISA speed of sound at 15 km is $a(h_0) = 295.1\,\mathrm{m\,s^{-1}}$, produces a launch velocity of:

$$V_0 = M_0 \cdot a(h_0) = 2 \times 295.1 = 590.1\ \mathrm{m\,s^{-1}} \tag{30}$$

The potential energy contribution is:

$$\Delta V_{\text{pot}} = \Delta V_{\text{ground}} - \sqrt{2\left|\varepsilon_{\text{orb}} + \frac{\mu}{R_E + 15{,}000}\right|} \tag{31}$$

$$\Delta V_{\text{pot}} = 8021 - 8002.7 = 18.3\ \mathrm{m\,s^{-1}} \tag{32}$$

The total ΔV gain is therefore:

$$\Delta V_{\text{gain}} = \Delta V_{\text{pot}} + V_0 = 18.3 + 590.1 = 608.4\ \mathrm{m\,s^{-1}} \tag{33}$$

The mass ratios are:

$$\frac{m_0}{m_f} = e^{\Delta V/v_e} = \begin{cases} e^{8021/3433} = 10.35 & \text{ground launch} \\ e^{7413/3433} = 8.63 & \text{air launch} \end{cases} \tag{34}$$

where $v_e = I_{\text{sp}} \cdot g_0 = 3433\,\mathrm{m\,s^{-1}}$ for a liquid-propellant engine with $I_{\text{sp}} = 350\,\mathrm{s}$. The corresponding propellant mass fractions are:

$$f_p = 1 - \frac{1}{m_0/m_f} = \begin{cases} 90.3\% & \text{ground launch} \\ 88.4\% & \text{air launch} \end{cases} \tag{35}$$

Table 5: Comparison of ground and proposed air-launch scenario for a single-stage rocket.

	Ground launch	Air launch
ΔV required [$\mathrm{m\,s^{-1}}$]	8021	7413
Mass ratio m_0/m_f [-]	10.35	8.63
Propellant fraction [%]	90.3	88.4
Structure + payload fraction [%]	9.7	11.6

In conclusion, the proposed scenario of $h_0 = 15\,\mathrm{km}$ and $M_0 = 2$ satisfies both feasibility and efficiency criteria for a single-stage rocket mission. The Mach 2 launch velocity is achievable by an existing supersonic military aircraft at a flight-path angle of 30°, while the 26% reduction in atmospheric density at 15 km lowers all major loss terms. The resulting ΔV_{gain} of $608.4\,\mathrm{m\,s^{-1}}$ reduces the mass ratio from 10.35 to 8.63 and increases the structure and payload fraction from 9.7% to 11.6%, making the single-stage mission significantly more achievable.

Code

The code for this assignment is available on GitHub

References

[1] N. Sarigul-Klijn, M. Sarigul-Klijn, C. Noel, "Air-Launching Earth to Orbit," *J. Spacecraft Rockets*, vol. 42, no. 3, pp. 569–572, 2005.